

A new multi-period investment strategies method based on evolutionary algorithms

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Abstract This work introduces a new algorithmic trading method based on evolutionary algorithms and portfolio theory. The limitations of traditional portfolio theory are overcome using a multi-period definition of the problem. The model allows the inclusion of dynamic restrictions like transaction costs, portfolio unbalance, and inflation. A Monte Carlo method is proposed to handle these types of restrictions. The investment strategies method is introduced to make trading decisions based on the investor's preference and the current state of the market. Preference is determined using heuristics instead of theoretical utility functions. The method was tested using real data from the Mexican market. The method was compared against buy-and-holds and single-period portfolios for metrics like the maximum loss, expected return, risk, the Sharpe's ratio, and others. The results indicate investment strategies perform trading with less risk than other methods. Single-period methods attained the lowest performance in the experiments due to their high transaction costs. The conclusion was investment decisions that are improved when information providing from many different sources is considered. Also, profitable decisions are the result of a careful balance between action (transaction) and inaction (buy-and-hold).

Keywords Evolutionary algorithms · Portfolio optimization · Algorithmic trading · Multi-objective

1 Introduction

Financial investment is an important economic activity. For example, the New York stock exchange (NYSE) had an estimated monthly value of \$1.55 USD trillions and a year-to-date value of \$14.72 USD trillions as of December 2015 [9]. The value of indexes like the Dow Jones Industrial Average (DJI), the Standard & Poor 500 (S&P500), or domestic stock market indexes are common measure of a country's development. Financial markets provide a comfortable method to generate profit from diverse industries and commercial activities without direct involving of investors in a company's operation.

Investors are always concerned with the outcome of their decisions. Our inability to predict the future state of the market implies that risk is associated with any investor action. *Algorithmic trading* is the use of computational methods to assist investors in this task. Evolutionary algorithms (EAs) have been proposed to implement trading systems [1]. They are preferred because of their ability to handle restrictions that are difficult for other methods.

Traditional finance theory considers three main approaches for trading: fundamental analysis methods, technical analysis methods, and portfolio-based methods [25]. Fundamental analysis methods are focused on the valuation of securities. The accounting information from a company is used to estimate the true value of their stocks. Mis-priced securities will certainly reach their true value in the near future. Therefore, their early identification allows devising profitable strategies from this knowledge (e.g., buying low to sell high). Technical analysis holds that all

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the relevant information about a stock is contained in its data history. Technical analysis methods search for patterns in data and exploit them to find profit opportunities. Chart analysis, Bollinger's bands, and other related techniques can be found in the literature [17].

The third approach is portfolio-based methods. Markowitz [20] made a breakthrough in investment theory with the concept of *diversification*. This approach quantifies the uncertainty about the outcome of a decision (i.e., *risk*) and includes it into the optimization process. Optimal decisions are those that have maximum expected return and minimum risk. Diversification holds a set of carefully selected assets (i.e., the portfolio), which provides investors with the ability to control the level of risk of their decisions. Our approach is based on portfolios due to their risk management capabilities.

Figure 1 illustrates the main points of traditional portfolio theory. The stars represent the risk and return of a particular stock. The solid line is the Pareto front of optimal portfolios, which are combinations from these assets. Figure 1 shows that portfolios attain higher return with less risk than stocks. In other words, portfolios *dominate* individual assets. The highest return stock is always part of the Pareto front. The dashed lines are called *indifference curves*. They are a mapping from the investor's utility function for combinations of risk/return with the same level of utility. Portfolio theory holds that there is only one single point where the indifference curves and the Pareto front touch each other tangentially. The corresponding portfolio is the one which satisfies the investor's preferences about risk and return.

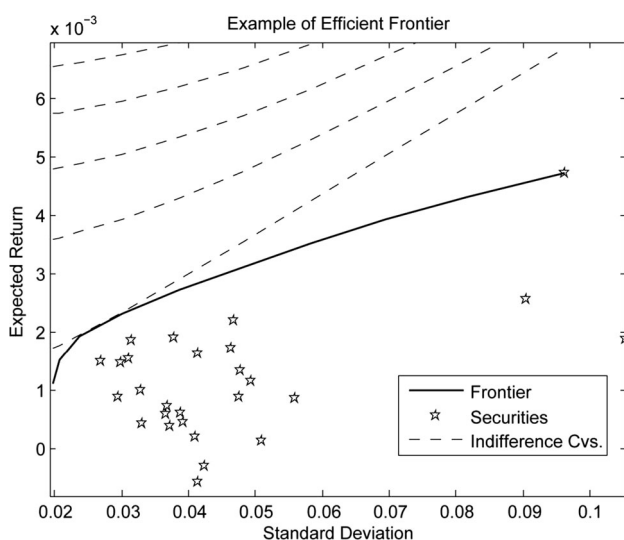


Fig. 1 Efficient frontier, individual securities, and indifference curves

1.1 Limitations of traditional approaches

The original portfolio optimization problem is solved using quadratic programming. The critical line method (CLM) was developed to find optimal portfolios in an efficient way [21]. Although the limitations of this approach decimate its performance in real-world investment, the inclusion of restrictions, like rounded lots, cardinality, compulsory assets, or transaction costs, prevents the direct application of traditional solution methods. EAs have been used to tackle the problem of portfolio optimization with restrictions. For example, Lwin, Qu, and Kendall [18] studied a traditional portfolio selection problem with cardinality, quantity, compulsory assignment, and round-lots restrictions. That work proposed a new multi-objective evolutionary algorithm (MOEA) for the task and compared it against the non-dominated sorting genetic algorithm (NSGA-II), the Pareto envelope selection algorithm (PESA-II), the strength Pareto evolutionary algorithm (SPEA-2), and the Pareto archived evolutionary strategy (PAES). Gupta, Mehlaawat, and Mittal applied genetic algorithms (GAs) and support vector machines (SVM) to a similar problem [11].

Some works have studied alternative risk metrics. For example, Ranković [24] proposed using evolutionary algorithms to solve portfolio selection problems with value-at-risk (VaR) instead of standard deviation to quantify risk. Value-at-risk can be understood as the q -quantile of the distribution of the portfolio's return. In other words, VaR indicates the expected value of loss for a given probability q , which is usually small. Their method was implemented with the SPEA-2. Other risk metrics, like the weighted expected shortfall (WES), were studied in other works [28].

On the other hand, finance theory had acknowledged the limitations of the Markowitz's approach since early references [5]. Their efforts have focused their attention on *dynamic restrictions* where transaction costs are the most prominent one. It refers to fees paid to brokers to execute the buy/sell orders from the investor. In general, dynamic restrictions are those related to the change of the portfolio with time. Chen [5] devised a multi-period definition of the problem where the model considered the portfolio will be liquidated at final time. Therefore, there are T decisions (i.e., portfolio's changes) to be optimized. This means the optimal solutions are sets of T portfolios to be applied at each period. The decision is made at time 0 with the current information about the stock's returns. Transaction costs and other dynamic restrictions are easily introduced in the problem definition, which was derived using dynamic programming techniques. Further research concluded single-period problems (i.e., Markowitz's definition) are different from multi-period portfolio problems when the

utility function of the investor depends on the state of the market [8].

The multi-period portfolio problem proved to be difficult to solve. Chen [5] did not provide a solution to it. Later references introduced the definition of auxiliary problems to effectively reduce the feasible portfolios and provide a solution [16]. This approach became standard in later references. For example, Çanakoğlu, Özekici [3] studied multi-period problems with transaction costs and exponential utility functions. Also, Yi [29] proposed a solution for multi-period models with transaction costs and a quadratic utility function. Besides, these references exploited some properties of the utility function or the risk metric to solve the problem. These references focused on finding closed-form solutions to it. Other works presented comparative studies between variance and conditional value-at-risk metrics for multi-period problem definitions [4].

On the other hand, the application of evolutionary algorithms to multi-period portfolio problems has received limited attention in the literature. One example is Zhang's report [30]. That work defined the problem to be a case of stochastic programming, where the possible states of the market are represented using trees [12]. A genetic algorithm was used to solve a multi-period portfolio optimization problem which includes transaction costs and conditional VaR.

1.2 Limitations of multi-period approaches

Previous references regarding multi-period portfolios are mainly concerned with finding closed-form solutions to the problem for a given combination of risk metrics and utility functions. They overlook numerical solutions, which are usually acceptable in practical applications. Besides, they do not consider two factors, which could make the approach more practical: the utility of other periods besides the last one, and data innovations. The first point refers to the fact that current multi-period approaches only consider sets of a fixed number of portfolios to be optimized. On the contrary, our approach considers sets of a variable number of portfolios to be optimized. They could provide the investor with a wider range of investment options if they truly exist, which is the case when sets of a variable number of portfolios are part of the Pareto front. Proof about their existence is provided in the results section.

The second point refers to current multi-period approaches which ignore the arriving data during the investment time because the decision is made beforehand, based on historic data. That approach assumes the distribution of returns will remain stationary along the investment horizon. Nevertheless, real-time portfolio evaluation

could lead to corrective actions when the current performance indicates the final outcome will not abide to the investor's expectations. This point is discussed in later sections.

Another limitation is that most of the studied solution methods rely on theoretical utility functions to model the investor's preference. They assume all the investors will have the same behavior. On the other hand, an identification process could determine the preference of a particular investor in the form of suitable parameters to be included into the optimization process. Identification is a common practice in other fields (e.g., control engineering [31]).

Finally, other researchers usually consider transaction costs but overlook other dynamic restrictions. Our work extends the subject by including portfolio unbalance and inflation into the problem definition.

1.3 Overview

The rest of this paper is divided in the following order. Section 2 describes a MOEA to solve multi-period portfolio problems with transaction costs, unbalance, inflation and inclusion of data innovations. Besides, the Pareto front may include portfolio sets with heterogeneous number of periods. Section 3 introduces the investment strategies (ISs) method. It allows the application of the algorithm to make investment decisions based on the investor's preference and the current market conditions. Section 4 describes the experiments. Section 5 presents the discussion of the results. Conclusions appear in Sect. 6.

2 Evolutionary algorithm for multi-period portfolio optimization

The first part of this section describes the objective function. It includes the problem definition with dynamic restrictions. The second part introduces the evolutionary algorithm.

2.1 Objective function

The *portfolio* $\mathbf{w}(t)$ is a vector of weights defined as

$$\mathbf{w}(t) = [w_1(t) \quad \dots \quad w_m(t)], \quad (1)$$

restricted to

$$\sum_{i=1}^m w_i(t) = 1, \quad (2)$$

where m is the number of assets in the portfolio. $\mathbf{w}(t)$ indicates the fraction of total wealth allocated to each stock.

Finance theory usually expresses the changes of price in the form of *rates of return*. $\mathbf{r}(t)$ is a row vector of returns, where

$$r_i(t) = \frac{p_i(t) - p_i(t-1)}{p_i(t-1)}. \quad (3)$$

In Eq. 3, $p_i(t)$ represents the price of the i th stock at time t . The current value of the portfolio $x(t)$ can be obtained from the current values of $\mathbf{w}(t)$, $\mathbf{r}(t)$, the interest rate I , and the past portfolio's value $x(t-1)$ with the following expression:

$$x(t) = \frac{\mathbf{r}_1(t)\mathbf{w}(t)^T}{(1+I)^t} (1 - \text{cost}(t))x(t-1), \quad (4)$$

where $\text{cost}(t)$ is defined as

$$\text{cost}(t) = c \left\| \mathbf{w}(t) - \frac{\mathbf{r}_1(t-1) \odot \mathbf{w}(t-1)}{\mathbf{r}_1(t-1)\mathbf{w}(t-1)^T} \right\|_1. \quad (5)$$

In expressions 4 and 5, $\mathbf{r}_1(t) = 1 + \mathbf{r}(t)$. $\text{cost}(t)$ depends on the transaction cost rate c , and the 1-norm between the new portfolio and the last unbalanced portfolio. The $\odot$ operator denotes element-wise matrix product. The interest rate I represents the effect of inflation, based on the concept of present value [25]. It holds the utility of profit decreases with time. For example, some people would prefer earning \$1000 today than earning \$10,000 in 20 years. I is part of a damping factor, which includes the change of utility with time.

In Eq. 5, unbalance depends on the past returns $\mathbf{r}(t-1)$ and the past portfolio $\mathbf{w}(t-1)$. Unbalance is the natural change in the portfolio composition due to differences in the return rates of assets. For example, if a 2-assets portfolio has the same amount of wealth invested in each one of them, the portfolio weights are $\mathbf{w} = [0.5, 0.5]$. Let us suppose the returns for each asset are $\mathbf{r}_1 = [0.9, 1.2]$ and total wealth to be $x = 1$. Ignoring dynamic effects, the amount of wealth of the first security will decrease, and the second one will increase to be $[0.45, 0.6]$. The new portfolio weights are $\mathbf{w} = [0.4286, 0.5714]$. Unbalanced portfolios tend to accumulate wealth in the most profitable security while increasing the portfolio's risk. Financial professionals usually recommend portfolio revision to correct unbalance and keep risk under desired levels [25].

Equation 4 can be applied iteratively to obtain the portfolio value at any time period desired. Our approach considers $t = 1, 2, \dots, T$. The probability distribution of the portfolio's value can be estimated using a Monte Carlo approach. In this case, several random scenarios of returns $\mathbf{r}(t)$ are generated from a multivariate normal distribution $\mathbf{r}(t) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. (6)

Nevertheless, the proposed approach is not restricted to any particular distribution. In Eq. 6, $\boldsymbol{\mu}$ is the vector of the average returns of securities, and $\boldsymbol{\Sigma}$ is the covariance matrix of securities. Both variables are computed from the available data history. The optimization objectives are the expected value of the portfolio's wealth $E[x(T)]$ and its risk $\sigma(T)$. Both values are estimated from the evaluation of final return of the set of portfolios using the random realizations and Eq. 4. This approach allows the application of different risk metrics. In this work, both standard deviation and VaR were studied in the experiments.

2.2 Multi-objective risk evolutionary algorithm

This work proposes a new multi-objective algorithm to solve multi-period portfolio problems. A general definition of multi-objective problems can be found in some references [23]:

$$\min f_m(\mathbf{x}), \quad m = 1, 2, \dots, M; \quad (7)$$

subject to

$$g_j(\mathbf{x}) \leq 0, \quad j = 1, 2, \dots, J; \quad (8)$$

$$h_k(\mathbf{x}) = 0, \quad k = 1, 2, \dots, K; \quad (9)$$

where $\mathbf{x}$ is the vector of decision variables, f_m are the objective functions, and g_j and h_k are the problem constraints. In Eq. 7, minimization of all the objectives is assumed without loss of generality.

Given Eqs. 7 to 9, a vector $\mathbf{x}$ *dominates* other decision vector $\mathbf{y}$ when $f_m(\mathbf{x}) \leq f_m(\mathbf{y})$ for all the objectives and $\mathbf{x} \neq \mathbf{y}$. Domination is widely used in MOEAs to favor promising solutions. Given the space of feasible solutions $\mathcal{F}$, $\mathbf{x}$ is *Pareto-Optimal* when there is no other vector $\mathbf{y} \in \mathcal{F}$ which dominates $\mathbf{x}$. The set of Pareto-optimal solutions is called the *Pareto front*.

The structure of some popular multi-objective algorithms, for example the NSGA-II [7], suggests that this type of algorithms is composed by the following elements: A mechanism to preserve the best solutions, a mechanism to favor solutions closer to the Pareto front, and a method to favor solutions located at low-populated sections of the solution space. The NSGA-II accomplishes these tasks using elitism, non-domination sorting, and crowding, respectively. Some algorithms like the PAES [14] or the PESA [6] rely on external archives to save the non-dominated solutions along the run. Moreover, other works have proposed using non-generational evolutionary algorithms for solving multi-objective problems [27]. Our approach also favors a non-generational scheme because it naturally works like an archive, avoiding the problems of other algorithms based on external archiving.

2.2.1 Area metric (A_{ms})

In our method, the closeness of the individuals to the non-dominated front is estimated using the *area metric*. The concept is described in Fig. 2. The solid line represents the non-dominated front, and the points are solutions. The term *non-dominated front* is used to indicate this one is not necessarily the true Pareto front. Non-dominated solutions are labeled by Roman numbers, and dominated solutions are labeled using capital letters. The shaded areas denote the region dominated by a particular solution. Darker tones indicate that particular area is dominated by more individuals than clear toned areas. In Fig. 2, we can observe the areas close to the non-dominated front have clearer shades than the ones far away from it. This occurs because areas covered by non-dominated individuals overlap more frequently. The information provided by the position of non-dominated individuals can be used to guide the optimization process. Individuals located at clear-shaded areas are favored because this indicates they are closer to the non-dominated front.

In this example, *C* is the best solution because it is only dominated by IV. *A* is dominated by I and II, while *D* is dominated by IV and V. *B* is dominated by I, II, and III. *E* is dominated by all of them. *A* and *D* would have the same fitness value under this criterion.

Also, the spread of individuals is promoted by this method. Areas under highly populated sections of the non-dominated front will present more overlapping. For example, if II was not present, then both *A* and *C* would be dominated

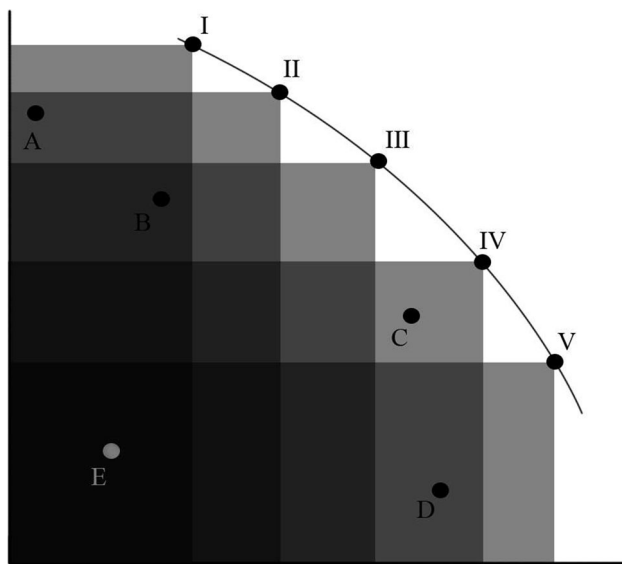


Fig. 2 Structure of Pareto front and dominated individuals

by only one solution. Besides, both *B* and *D* would be dominated by two solutions. In this case, both *A* and *B* were benefited because they were covered by a less populated section of the non-dominated front. The method assumes the empty sections will be covered by the offspring of the favored individuals. Similar underlying assumptions are common in other multi-objective algorithms.

In the example, both *A* and *D* have the same fitness value. A method merely based on the number of dominating individuals could have a limited resolution. To overcome this problem, the area covered by the individuals is used in the calculation of A_{ms} . In this case, *D* covers a larger area of the solution space than *A*, then *D* would have a higher fitness value. Therefore, an expression for the proposed metric is the following:

$$A_{ms}(\mathbf{J}_k) = \begin{cases} \frac{\prod_{i=1}^{\ell} \mathbf{J}_k(i)}{\prod_{j=1}^{Nd_k} \prod_{i=1}^{\ell} \mathbf{J}_j(i)}, & \text{if } Nd_k > 0 \\ 1, & \text{if } Nd_k = 0. \end{cases} \quad (10)$$

In this work, the individuals from the non-dominated front which dominate a particular solution will be called its *dominant*. In Eq. 10, $\mathbf{J}_k$ is the vector of evaluations of the k th individual, and ℓ is the number of objectives. Nd_k is the number of dominants of the k th individual from the population. $\mathbf{J}_j$ is the vector of evaluations of the j th dominant of individual k . Equation 10 is the ratio of the area (or hyper-volume) covered by the dominated individual and the areas of their dominants. A non-dominated individual has $A_{ms} = 1$. Moreover, duplicated, non-dominated individuals receive an A_{ms} value inversely proportional to the number of existing copies in the population.

2.2.2 Density metric (C_{ms})

A_{ms} is useful to favor individuals located at low-populated areas close to the non-dominated front. Nevertheless, the information provided by A_{ms} decreases when the population becomes non-dominated, which occurs at the last stages of the run. Density metrics (e.g., crowding) become prominent at that point because they are the only remaining source of information to guide the search for optimal solutions. Köppen and Yoshida [15] have proposed alternatives to crowding to enhance the search power of multi-objective algorithms when many objectives (more than 3) are optimized. A lack of dominated solutions might occur at early stages of the run in a many-objectives problem. From this reference, our work implements the *negative ε -domination* metric to favor individuals located at low-populated sections of the solution space. The procedure is described in the following pseudocode:

Algorithm 1 Negative ε -Domination

```

function EPSD( $J_{NL}, k, N, L$ )
   $eps = \infty * ones(1, N)$ 
  for  $i = 1 : N$  do
    for  $j = 1 : L$  do
      if  $i \neq k$  then
         $eps(i) = \max(eps(i), |J_{NL}(i, j) - J_{NL}(k, j)|)$ 
      end if
    end for
  end for
  return  $\min(eps)$ 
end function

```

This algorithm computes C_{ms} as the minimum of the maximum differences from the i th individual to the rest of the population for each objective. Normalizing is used to avoid scaling problems. C_{ms} is computed for the non-dominated individuals at each cycle, while dominated individuals have $C_{ms} = 0$. These calculations are computed in the function above. Finally, the fitness of the k th individual f_k is defined to be the sum of both metrics

$$f_k = A_{ms}(k) + C_{ms}(k). \quad (11)$$

2.3 An example

The example is defined for maximization of all the objectives. This allows using the coordinates axis to define the areas covered by the individuals and makes them easier to visualize. This change can be done without loss of generality. In Fig. 3, the non-dominated individuals are denoted by Roman numbers. The coordinates of each individual are the following: $A = [1, 4]$, $I = [2, 6]$, $II = [5, 5]$, and $III = [6, 3]$. $A_{ms}(I) = A_{ms}(II) = A_{ms}(III) = 1$ because they are the non-dominated individuals from the

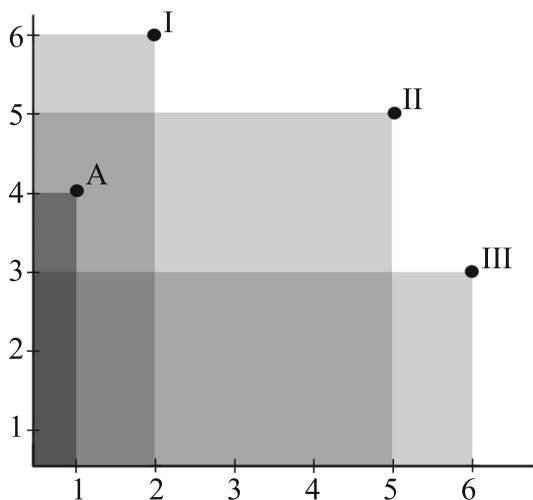


Fig. 3 Example of the application of area measure and crowding measure

population. Equation 10 is applied to compute $A_{ms}(A)$. In this case, $\text{Area}(A) = 1 \times 4 = 4$, $\text{Area}(I) = 2 \times 6 = 12$, and $\text{Area}(II) = 5 \times 5 = 25$. $\text{Area}(III)$ is not necessary because $III \not\prec A$. Therefore, $A_{ms}(A) = \frac{4}{12 \times 25} = \frac{1}{75}$.

On the other hand, $C_{ms}(A) = 0$ because A is a dominated individual. The difference between individuals is defined as $\Delta_{a-b} = |J_a - J_b|$. Therefore, $\Delta_{I-II} = |[2, 6] - [5, 5]| = [3, 1]$, $\Delta_{I-III} = |[2, 6] - [6, 3]| = [4, 3]$, and $\Delta_{II-III} = |[5, 5] - [6, 3]| = [1, 2]$. In a similar fashion, $\Delta_{I-A} = [1, 2]$, $\Delta_{II-A} = [4, 1]$, and $\Delta_{III-A} = [5, 1]$. For individual I, the relevant differences are Δ_{I-II} , Δ_{I-III} , and Δ_{I-A} . The maximum differences for each objective are computed, and the result is $[4, 3]$. The minimum from the vector is $-\varepsilon_I = \min[4, 3] = 3$. The values $-\varepsilon_{II} = 2$ and $-\varepsilon_{III} = 3$ are found in the same manner. These values are finally normalized based on the maximum $-\varepsilon$ from the population; therefore, $C_{ms}(I) = C_{ms}(III) = 1$, and $C_{ms}(II) = \frac{2}{3}$.

2.4 Encoding of portfolios

The chromosome is divided into two parts. The first one is used to encode the portfolio, while the last one contains the revision time. The portfolio will hold the number of periods indicated by the chromosome; it will drift freely until revision time to avoid transaction costs. This approach allows the simultaneous optimization of portfolios with different number of periods. Besides, implementation using virtual genes [26] is specially useful to avoid redundancy when the number of maximum periods is not 2^n .

The coding uses $m - 1$ genes to encode the portfolio. This approach visualizes portfolio weights as the components of a unity vector. It takes the advantage the sum of the squares of the components of a unity vector sums 1; the same occurs with portfolio's weights. Genes encode the angles between the unity vector and each plane of the coordinates system. The weight w_i is defined by the following equation:

$$w_i = (c_i \sin \theta_i)^2, \quad (12)$$

where θ_i is defined as

$$\theta_i = \left(\frac{\pi}{2}\right) \left(\frac{g_i}{2^b - 1}\right). \quad (13)$$

Equation 12 represents the i th portfolio weight from $i = 1, 2, \dots, m - 1$. g_i is the decimal value represented by the i th gene. In Eq. 13, g_i is normalized using the maximum count possible with b bits and converted to radians to obtain θ_i . Figure 4 shows θ_i is the complement of the angle formed between the vector and the corresponding axis. The weights are computed from the component normal to the plane used to define θ_i . c_i is the magnitude of the projection of the vector on the plane used to define θ_i . In the

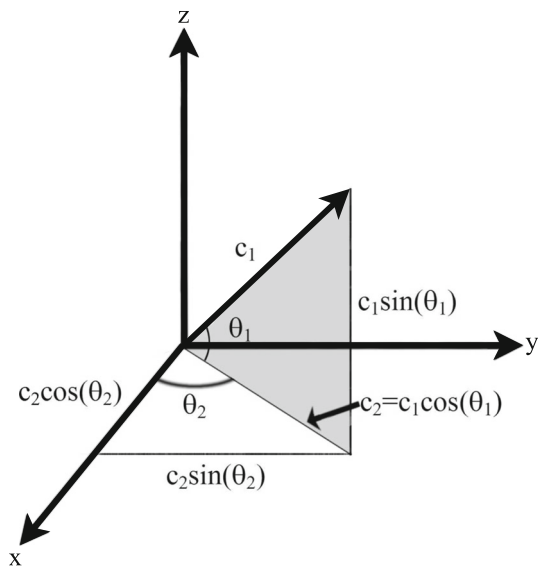


Fig. 4 Graphical description of portfolio coding. The components of vector c are computed iteratively. The last component is the last projection c_{M-1} . The portfolio weights are obtained by *squaring* the component values

beginning, $c_1 = 1$ because $\mathbf{C}$ is an unity vector. The next projection is computed from the following equation:

$$c_{i+1} = c_i \cos(\theta_i). \quad (14)$$

The last weight is the squared value of the last projection:

$$w_m = (c_{m-1} \cos(\theta_{m-1}))^2. \quad (15)$$

3 Algorithmic trading method

The proposed algorithm is used to optimize multi-period sets of portfolios with a given number of periods T . It outputs an estimated Pareto front of multi-period portfolios with different levels of expected return and risk. The inputs are the data history, a given initial wealth x_0 , and initial portfolio $\mathbf{W}_0$. It is used to implement the investment strategies trading method (ISs) explained below.

The first step of the method is identification of the investor's preference of expected return and risk. The algorithm is used to find the non-dominated front of optimal portfolios for $T = 1$ and out-of-market initial conditions (i.e., no wealth allocated to any asset). If the expected return of the portfolio with the highest risk is not enough for the investor, the algorithm is run again for $T = 2$. The non-dominated front will be the combination of the fronts with $T = 1$ and $T = 2$, discarding dominated solutions. This process is repeated for $T = 3, 4, \dots$ until the level of risk of the highest return portfolio is unbearable for the

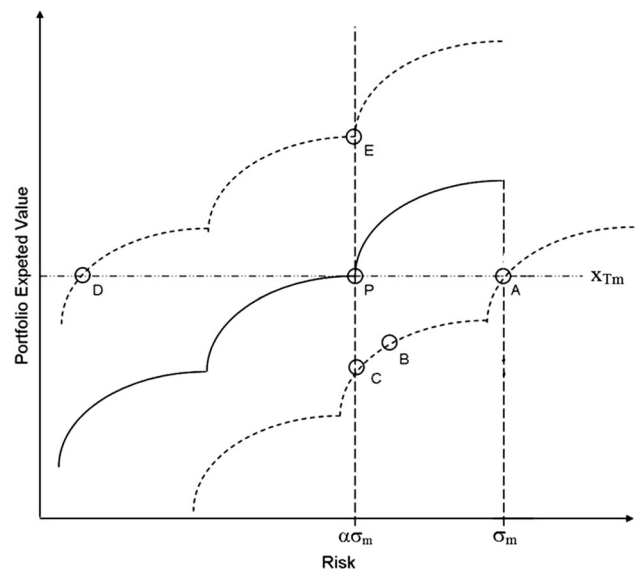


Fig. 5 Multi-period Pareto fronts. The *solid front* is computed for coming into market conditions. *Dashed fronts* illustrate the cases when the decisions were profitable and when decisions caused a loss. Point P represents the initial portfolios set chosen by the investor

investor. The method is illustrated in Fig. 5. The solid line represents the resulting multi-period Pareto front. Its maximum risk level is labeled σ_m . The parameter α is used to express the fraction of the maximum risk the investor is willing to take, based on the output. Therefore, the portfolio P which satisfies the investor's preference has risk $\alpha\sigma_m$ and expected return x_{Tm} . In the rest of this work, the vector $[x_{Tm}, \alpha\sigma_m]$ is considered the *goal* of the optimization. This means the algorithm will run until the portfolio attains this level of return and will try to avoid higher risks levels than $\alpha\sigma_m$.

The algorithm applies the selected portfolio for the specified number of periods. Then, the portfolio is evaluated using the new data obtained from the market. The algorithm computes the new portfolio $\mathbf{W}_T$ and its value x_T . The run will continue while $x_T < x_{Tm}$. Also, the history of data is updated with the innovations obtained during the time the portfolio was held. A sliding window approach is used to discard old data. This is necessary due to the non-stationary properties of the market.

In Fig. 5, the dashed lines represent the multi-period Pareto fronts obtained for two different scenarios. The one above the solid line is the case when $x_T > x_0$. The one below corresponds to the loss scenario. The profit scenario provides the investor with several options, which are better than the goal. On the other hand, the loss scenario shows the investor must take higher risk levels than $\alpha\sigma_m$ to reach x_{Tm} . A and D are the portfolios which satisfy the investor's preference about return, while C and E are the ones which satisfy his preference about risk. B and E are the portfolios

with closer risk and return values to the goal for their respective scenarios.

Our approach chooses portfolios according with the goal set by the algorithm. The selected portfolio from the current Pareto front is the one whose weighted distance to the goal is the minimum. The weighted distance Δ is defined to be

$$\Delta = \left\| \left([x_{Tm}, \alpha\sigma_m] - [x_j^1, \sigma_j^1] \right) \odot \mathbf{S} \right\|_2, \quad (16)$$

where x_j^1 and σ_j^1 are the expected return and risk of the j th individual from the current multi-period Pareto front. The norm used in Eq. 16 is the Euclidean distance. The *investment strategy* is a vector of weights defined as

$$\mathbf{S} = [s_1, s_2, \dots, s_\ell], \sum_i s_i = 1, \quad (17)$$

where ℓ is the number of objectives. The weights of $\mathbf{S}$ have a direct impact on the algorithm's behavior. For example, $\mathbf{S} = [1, 0]$ is similar to a Martingale gamble strategy [10]. A Martingale strategy considers the scenario of repeating gambling games. The gambler initially sets a goal and bets the necessary amount of money to reach it. He quits the game when the goal is reached. On the contrary case, the gambler bets the necessary amount to reach the goal and recover from his losses. The experiments study the behavior of different investment strategies. The algorithm chooses the next set of portfolios from the current multi-period Pareto front using the distance metric defined by Eq. 17. The set of portfolios with the minimum distance from the goal is selected by it. The algorithm runs while there are innovations available until the goal is reached.

4 Description of experiments

Weekly data from the Mexican Índice de Precios y Cotizaciones (IPC) from 2012 to 2014 were selected for the experiments. The data can be found online [13]. Random starting dates are selected for each experiment. From that point, the algorithm managed the portfolio to reach the goal. The goal was set from the first Pareto front, which was computed for periods from $T = 1$ to $T = 5$ and $\alpha = 0.8$. To select assets, the algorithm computed their expected return and standard deviation with the corresponding data, and the non-dominated stocks with positive rates of return were chosen for the optimization process. Besides, the interest rate I was computed to be the average return of the selected securities for the particular experiment. The transaction cost rate was set to be $c = 0.005$. A maximum number of periods $T_{\max}$ were given to the algorithm to reach the goal. The experiments considered a

time window $T_{\max} = 60$. Therefore, the algorithm discarded starting points unable to satisfy this restriction. Thirty random instances were chosen for the experiments.

The experiments tested different investment strategies. Five strategies with different levels of risk aversion were considered: $S_0 = [0, 1]$, $S_{25} = [0.25, 0.75]$, $S_{50} = [0.5, 0.5]$, $S_{75} = [0.75, 0.25]$, and $S_1 = [1, 0]$. Furthermore, the experiments tested two different risk metrics: the standard deviation and VaR of the final portfolio value. As mentioned above, VaR can be understood as the q -quantile of the distribution. Both risk metrics were applied to the same data sets to investigate their effect in performance. $q = 0.05$ in the VaR experiments.

The number of individuals in the population and the number of cycles are determined by Eq. 18:

$$n_{\text{inds}} = c_1 \lceil \log(T) \rceil, \quad (18)$$

where $c_1 = 50$ for the case of individuals, and $c_1 = 100$ for the case of cycles. The crossover probability was set to $p_c = 1$, and the mutation probability was $p_m = 0.1$. The maximum number of evaluations per individual was 10000. The number of bits per portfolio weight was $b = 4$.

Two other investment methods were tested for comparison purposes. The first one is buy-and-hold (B&H), which is widely used by investors. The same securities chosen by the algorithm were used for the B&Hs, but the portfolios are determined with the Markowitz's method (i.e., the critical line method); 10 portfolios were chosen from the Pareto front. The selected portfolios were held until the goal was reached or the time was over. These portfolios were not re-balanced during the experiment. In the results, S_1 denotes the lowest risk portfolio. B&H is relevant because it is an investment method that uses the minimum information possible in favor of avoiding transaction costs. The second method is a single-period investment method where the optimal portfolio was computed at each period with the Markowitz's method. Ten portfolios were selected from the efficient frontier for the test, where $M1$ denotes the lowest risk portfolio. This method is relevant because it uses the full information from data history, assuming that transaction costs and other dynamic restrictions may be ignored. In the experiments, 900 instances were evaluated considering the possible combinations of starting points, approaches, and risk metrics.

The same sets of random starting points were used to test the methods. This allowed a direct comparison of methods under the same conditions. Also, the overall performance of all the methods for a given instance is shown in later sections. The performance metrics considered were the following: maximum loss, total periods, final return, stop loss occurrence, domination at the risk/return plane, the Sharpe's ratio, and the average duration periods per

decision. Maximum loss refers to the minimum portfolio value reached during the experiment. Total periods is the time used by the algorithm to reach the goal. The total periods was equal to $T_{\max}$ when the algorithm was unable to reach the goal in time. The final return is usually close to the goal because the algorithm stops when the goal is reached. Nevertheless, this metric is useful to analyze the cases when a method failed the task. Stop loss time refers to the period when the portfolio wealth dropped below a certain value. In the experiments, a stop loss occurrence is counted when the portfolio's value drops below 70% of the initial wealth. Investors usually apply stop loss to prevent catastrophe, closing all the positions (i.e., selling the whole portfolio) when the stop loss signal takes place. The experiment was not aborted when a stop loss occurred, but it was registered to investigate the benefit of this practice. The Sharpe's ratio is a risk-weighted performance measure defined by the following equation:

$$Sh_p = \frac{\bar{r}_p - r_f}{\sigma_p}, \quad (19)$$

where r_p is portfolio return, r_f is the return of the risk-less asset, and σ_p is portfolio risk.

In finance, r_p and σ_p are usually computed from the portfolio price time-series. The problem of this approach is that estimations of return and volatility (therefore risk) are based on one single data realization only. Accurate volatility estimation is a more complex task. Some references have reported the use of the wavelet transform and GARCH models to estimate volatility [19]. In our approach, we take advantage of the proposed Monte Carlo method to estimate it. This approach computes an actual estimation of volatility from multiple realizations. The overall risk of the methods is estimated using Eq. 4 and the Monte Carlo approach. The risk and returns of the methods were plotted at the risk/return plane to investigate their domination relationships.

5 Discussion of results

The results appear from Figs. 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16. Figures 6, 7, 8, 9, 10 report the results grouped by method. For example, B&Hs from S_0 to S_1 , and single-period portfolios from M_0 to M_1 are grouped together, respectively. Figures 11, 12, 13, 14 show the results grouped by instance.

5.1 Box plots method

Box plots were chosen to present the results from the experiments. This method was preferred because it provides further information about the distribution of data. For example, box plots identify outliers clearly, which could

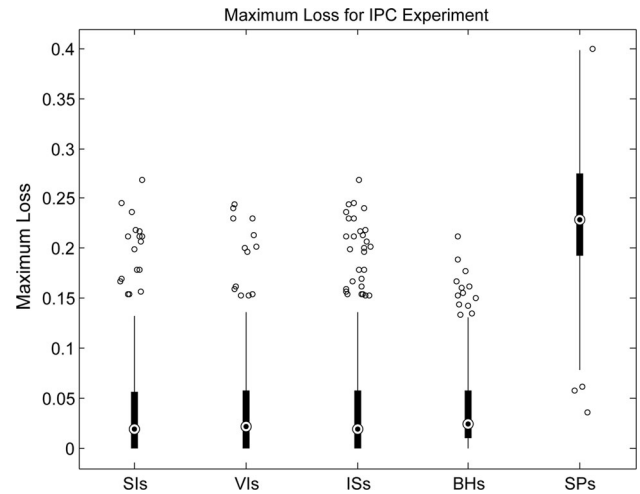


Fig. 6 Average maximum loss for IPC experiment per method

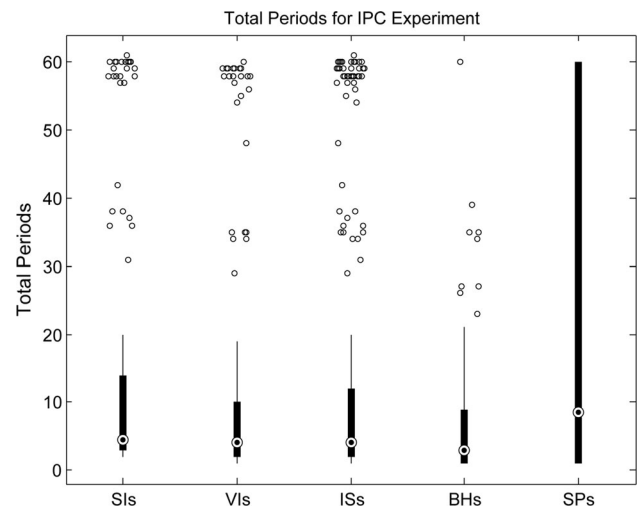


Fig. 7 Total periods for IPC experiment per method

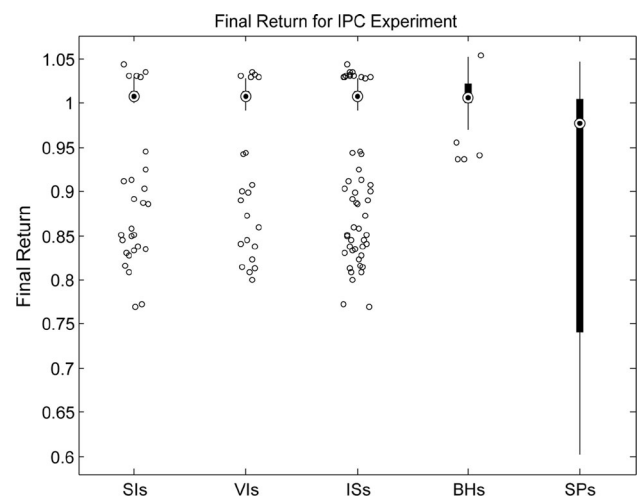


Fig. 8 Final return for IPC experiment per method

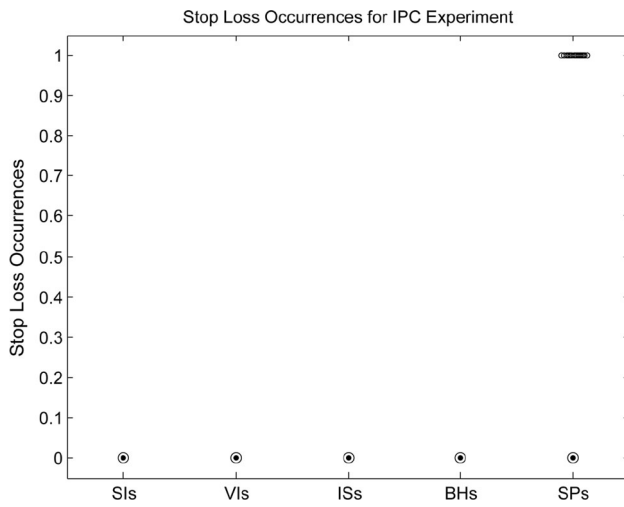


Fig. 9 Stop loss for IPC experiment per method

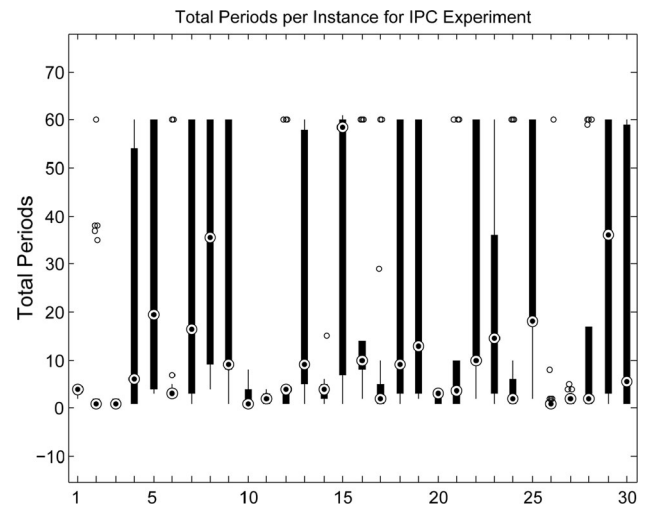


Fig. 12 Total periods for IPC experiment per instance

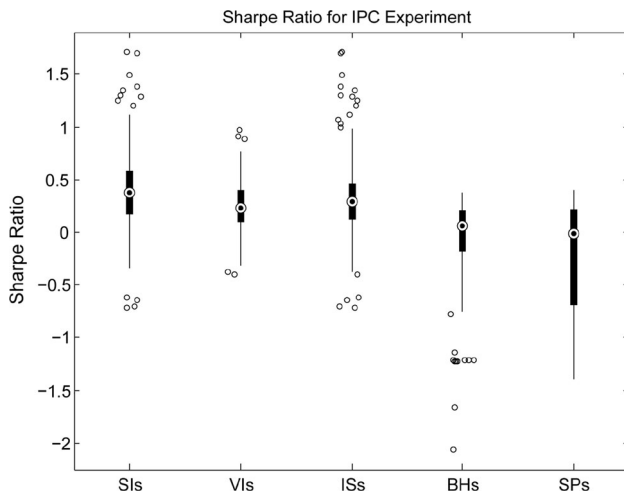


Fig. 10 Sharpe's ratio for IPC experiment per method

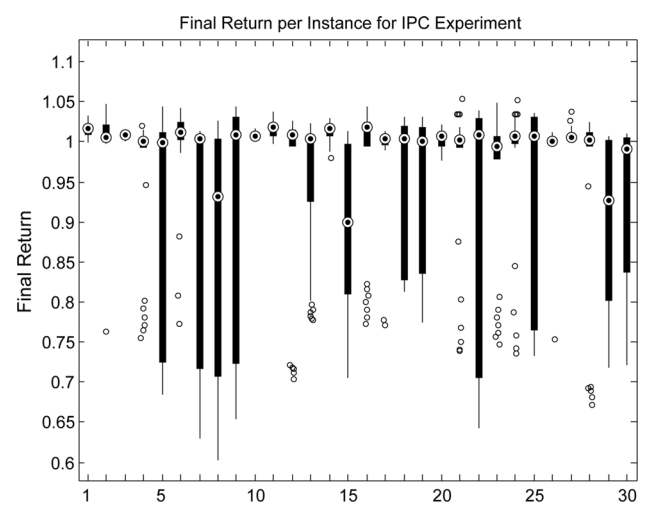


Fig. 13 Final return for IPC experiment per instance

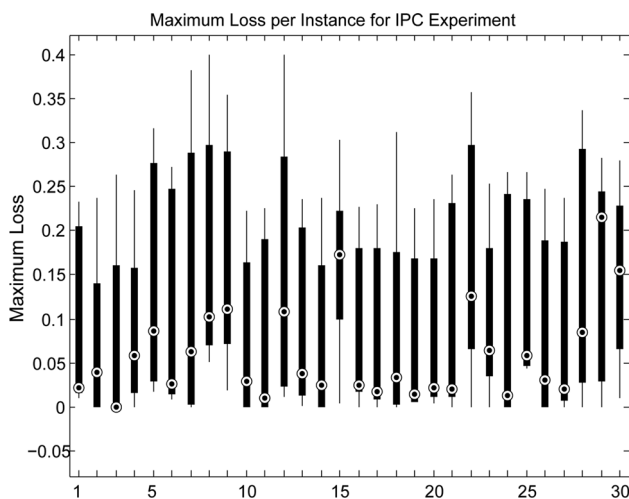


Fig. 11 Average maximum loss for IPC experiment per instance

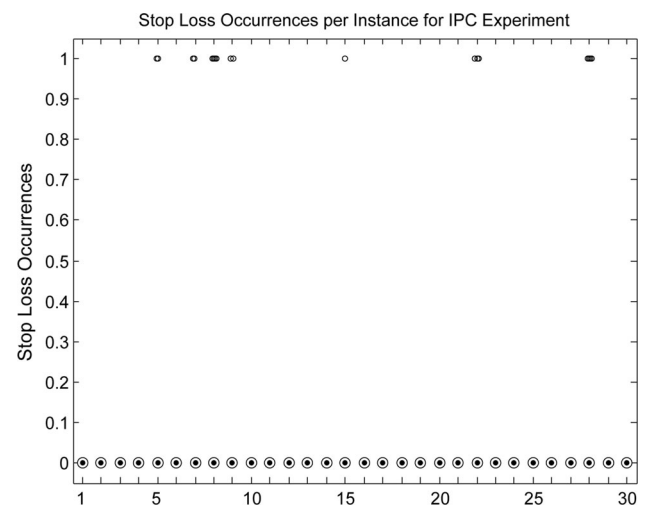


Fig. 14 Stop loss for IPC experiment per instance

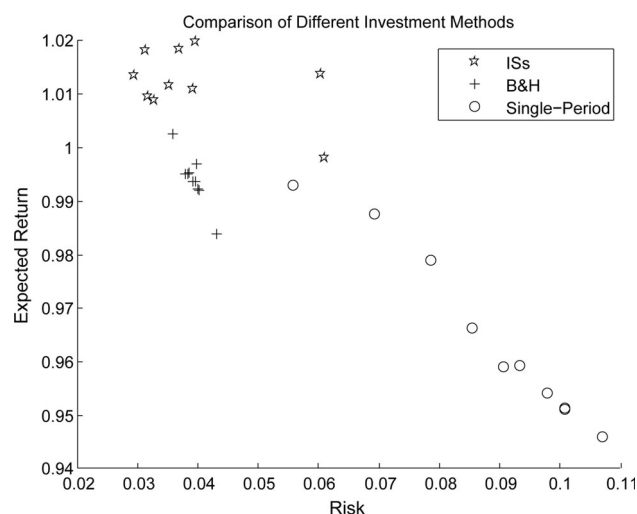


Fig. 15 Pareto front for IPC experiment

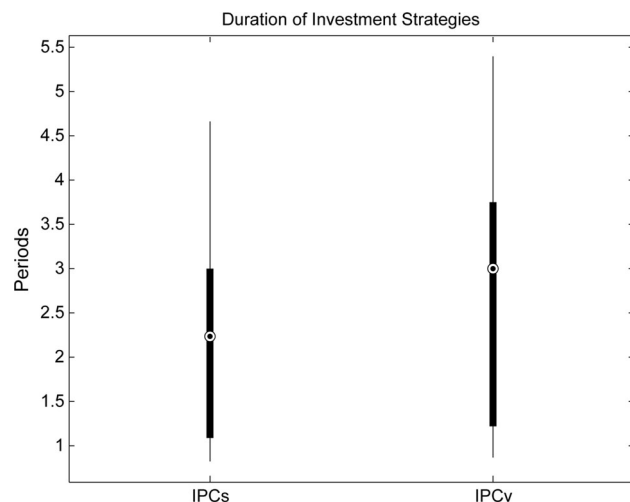


Fig. 16 Duration of different investment strategies

mislead results if they are not considered when interpreting them. The figures below show outliers were a common occurrence in the results; therefore, this method is useful in this case. The box represents the area from the first quartile to the third quartile of the distribution. The dot inside the box represents the median. The length of the whiskers indicates the minimum and maximum data, which are not considered outliers. The maximum whisker extension is $3/2$ of the size of the box. Data beyond those limits are considered outliers.

Hypothesis testing was conducted to investigate the significance of the conclusions obtained by the box plot method. The Kruskal–Wallis test is usually applied to compare the medians of different samples [22]. It is a nonparametric test, and this means it does not assume a particular data distribution. In general, the test explores whether the ranked samples come from the same

Table 1 Hypothesis test results for comparison of methods

IPC KW test	IS/BH	IS/SP	BH/SP	SI/VI
Maximum loss	0.0177	0.0000	0.0000	0.9738
Total periods	0.0000	0.0004	0.0000	0.0136
Final return	0.4940	0.0000	0.0000	0.8862
Sharpe's ratio	0.0000	0.0000	0.0140	0.0000
IPC χ^2 -GOF	IS	BH	SP	
Maximum loss	0.0000	0.0000	0.3248	
Total periods	0.0000	0.0000	0.0000	
Final return	0.0000	0.0000	0.0000	
Sharpe's ratio	0.0002	0.0000	0.0000	
KW test	IPCs/IPCv			
Duration	0.0003			
	IPCs	IPCv		
χ^2 -GOF	0.0000	0.0000		

distribution, which is the null hypothesis. Small p values (e.g., $p < 0.05$) indicate the evidence is enough to reject it. In that case, the comparison of medians provides useful information. Table 1 shows the results of the test. Also, the table presents results of the χ^2 goodness-of-fit (GOF) test to investigate the normality of data. This result is presented because other methods could be better than the Kruskal–Wallis test when data are normally distributed (e.g., one-way ANOVA). In Table 1, the label “KW test” indicates the results are the p values obtained from the Kruskal–Wallis test for the corresponding experiment. The methods are denoted in the following way: investment strategies (IS), buy-and-holds (BH), single-period portfolios (SP), investment strategies optimized using standard deviation (SI), and investment strategies optimized using value-at-risk (VI). The Kruskal–Wallis test can be applied to many samples simultaneously, but the results show the comparison of pairs of methods. For example, the label “IS/BH” indicates an investment strategies sample is compared against a buy-and-hold sample for the experiment of a particular performance metric. Table 1 shows the test results for every combination of metrics and pair of methods. Although the stop loss occurrence metric was not included, further testing was not necessary in that case. This is further discussed below.

In general, the p values from the χ^2 goodness-of-fit test indicate the data are not normal, which is somehow perceived in the box plots shown below. This justifies the use of the Kruskal–Wallis test. Besides, they indicate the maximum loss of single-period portfolios from the experiment (further explained below) could be normal. Nevertheless, the Kruskal–Wallis test was applied to all the cases.

Also, the results listed in Table 1 indicate most of the distributions of the results are statistically different from each other. This means the difference of medians is also significant and their comparison can be used to draw conclusions from the experiments. The cases where the null hypothesis was not rejected are discussed in the following sections.

5.2 Results

Figure 6 shows the maximum loss per method. The median of ISs is lower than the medians of B&Hs or SPs. The comparison between ISs and B&Hs is significant for a critical level of 0.05. On the other hand, there was not significant difference between SIs and VIs. The median of SPs is around 24%, while this level of loss was an outlier for the other methods. The presence of outliers is investigated in Fig. 11. The figure presents the box plots of maximum loss for each instance. This figure shows heavy losses (up to 40%) were observed in the experiments. The absence of outliers indicates heavy losses are explainable by the results. Besides, Figure 6 seems indicate heavy loss occurred when the methods were unable to reach the goal on time. Instances with higher median values seem to be harder. For example, instance number 29 has a median around 24%. This seems to indicate the outliers found in ISs and B&Hs are caused by this particular case. Probably those instance conditions were unsuitable for investing. Finally, the results indicate a stop loss higher than 40% would be ineffective for the studied data because it will never be triggered.

Figure 7 shows the total number of periods per method. SIs and VIs have similar performance, but SIs have a broader distribution bias to higher number of periods. B&Hs is the method with the lowest median and its followed by ISs. Cases with more than 20 periods were outliers for ISs and B&Hs, while these values were a common occurrence for SPs. Figure 12 indicates about 50% of the cases were easy for the methods. Having a number of periods larger than 10 periods is an outlier for those cases. The contrary occurred for the rest of the instances. Instance number 15 was the hardest one, where the median of the total number of periods was close to $T_{\max}$. This indicates the methods needed a longer time to reach the goal under those conditions. Nevertheless, the fact that the median is less than $T_{\max}$ indicates the methods were able to reach the goal despite the difficulties.

Figure 8 shows the final return per method. The termination of the run when the algorithm reached the goal avoided the occurrence of large, positive returns. The goal was selected based on the investor's preference, which is captured by the interest rate, the initial number of periods, and the value of $\alpha_{\sigma M}$. More ambitious goals would require a

revision of the investor's preference. These results are useful to investigate the cases where the methods were unable to reach the goal in time. SIs and VIs have similar median, but SIs distribution is biased toward positive returns, compared to VIs. The median of ISs is similar to B&Hs. Table 1 indicates there is not significant difference between both medians. Nevertheless, the differences are clear when comparing to SPs. The distribution of ISs is narrower than the distribution of B&Hs. This indicates the probability of having negative returns is higher for B&Hs than for ISs. SPs are clearly biased toward negative returns, where the lowest value was 60%. In the experiments, the range of final return is from 60% to 105%. There are some outliers in the results. This indicates there were cases where the methods were capable to surpass the goal. Nevertheless, further research is needed to draw conclusions about these fortunate cases. Figure 13 shows negative returns are likely for about 40% of the instances. These ones can be deemed to be difficult investment scenarios.

Figure 9 shows the stop loss occurrences for the IPC experiment. In this case, stop loss seldom occurred. Only SPs report occurrences, and they were outliers. Figure 14 confirms these results. Stop losses were outliers when they took place. Figure 14 indicates which cases presented stop loss occurrences. All the outliers are attributed to SPs. These results seem solid investment methods do not rely heavily on stop loss. However, they show they are useful when the method faces unexpected conditions. The stop loss value should be carefully selected or it will decimate the performance of the method. There is always the possibility of recovery when the portfolio is in the market, but it disappears when the positions are closed.

Figure 10 shows the Sharpe's ratio for the different methods. SIs have a significant larger Sharpe's ratio median than VIs and their distribution is biased toward higher values. ISs attained a significant higher median than B&Hs and SPs. This result suggests IS was able to attain a profit comparable to other methods, but with less risk. ISs distribution is more symmetrical than the others. Distribution of B&Hs and SPs is biased toward negative values, although the probability of negative values is higher for SPs than for B&Hs. Table 1 indicates the difference of medians between B&Hs and SPs is significant for a critical value of 0.05. On the other hand, the difference between VIs and SIs is also significant.

In general, Table 1 indicates there is no significant difference between SIs and VIs for any metric but the Sharpe's ratio. In that case SIs promised higher returns with lower risks than VIs. Standard deviation could be regarded as optimistic compared to VaR because it considers total risk instead of downside risk. In other words, investment strategies optimized using standard deviation consider the final outcome could be better or worse than

expected, while the ones optimized using VaR consider negative possible outcomes only. The difference between both metrics rely on the information they provide to the optimization process.

Figure 15 shows the Pareto front of the average risk–return for the different methods. The averages of risk and return of each method variation were plotted in the risk–return plane. The averages were computed along instances for each possible variation. For example, the risk and return of S_1 were computed for each instance, and their averages are plotted in Fig. 15. This figure shows the Pareto front is composed by ISs only. Besides, the variations of methods seemed clearly clustered together. Both ISs and B&Hs dominate SPs.

Finally, Fig. 16 compares the number of periods a portfolio was hold by the algorithm before revision. IPCs and IPCv denote the investment strategies applied to the IPC experiment optimized with standard deviation and value-at-risk, respectively. This figure indicates the algorithm applied sets of portfolios of different number of periods each time, which is a proof of the coexistence of them at the same multi-period Pareto front. The median of VaR ISs is higher than standard deviation ISs. Also, the distributions of VaR strategies are broader. This indicates the algorithm had a tendency to choose larger portfolios sets when VaR was considered as the risk measure. Table 1 confirms these results are significant. On the other hand, this is also a proof and overall risk metrics are relatively optimistic compared to downside risk metrics. In other words, the algorithm tends to be more cautious when VaR is considered by it. The algorithm seems to choose longer revision times to allow the portfolio to recover from unexpected losses. According to a Martingale strategy, a gambler would need an infinite amount of money to win every bet [10]. In a similar manner, a simple strategy to deal with loss is waiting for the market to recover by itself. Therefore, the number of periods of a set of portfolios seems to be related to the losses suffered during the run.

The results presented seem to indicate that ISs have better performance for the metrics presented, specially for the Sharpe's ratio and domination. The Sharpe's ratio results suggest ISs method provides comparable expected return with less risk than other methods. The maximum loss is significant for a critical value of 0.05, indicating that ISs had less drastic losses than B&Hs. This result seems to be correlated with the risk reduction of ISs. The least effective method was SPs, which indicates that neglecting transaction costs and other dynamic restrictions cannot produce optimal investment decisions. The outliers presented in the results seem to indicate that cases where heavy loss occurs are unavoidable regardless of the method used.

The effect of considering different risk metrics was investigated in this work. The results indicate VaR ISs have a tendency to generate larger sets of portfolios. This implies that risk increased more slowly with the number of periods when VaR was considered. In other words, the increase in risk from sets of portfolios with T periods to sets with $T + 1$ periods was more drastic when the standard deviation was considered instead of VaR. This could be from the fact that the probability distribution of the final return is not symmetric. VaR considers downside risk, while standard deviation considers total risk. Finance theory considers that the investor is not concerned by upward risk, which is regarded to be beneficial. Although no significant difference between the performances of standard deviations ISs and VaR ISs was found, the optimal portfolios found using each risk definition were different. The advantages of using different risk measurements should be further investigated.

Finally, it was found that the method truly used policies with different number of periods at each revision time. This is a proof that a multi-period Pareto front could be composed by sets of different number of portfolios. These Pareto fronts provide investors with more flexible options at the moment of decision making.

6 Conclusions

This work proposed a new trading method based on portfolio theory and evolutionary algorithms. A portfolio approach was preferred because it includes the risk of the decision into the optimization process. It acknowledges our inability to make a perfect prediction of the future and provides solutions considering the outcome could be worse than expected. From that perspective, portfolio optimization is similar to robust optimization, although there are differences which make risk optimization interesting on its own terms.

The solution model can be divided into two parts. The first part is the multi-objective evolutionary algorithm to solve multi-period portfolio problems. The second part is the investment method developed from the multi-period portfolio theory. The investment strategies method was developed to make investment decisions which considered dynamic restrictions like transaction costs, portfolio unbalance, and inflation.

The experiments showed this approach had better performance than buy-and-holds and single-period portfolios for the proposed metrics of performance. Besides, the experiments concluded the existence of multi-period Pareto fronts with heterogeneous time horizons.

The method devised in this work used the information provided by the data history in a different manner than

simple-period approaches. The multi-period model allowed the method to compute portfolio sets with different number of periods, while the single-period method makes a portfolio revision at each period. This means single-period methods use all the information available from the price data. Nevertheless, it attained the lowest performance in the experiments. On the other hand, the buy-and-hold method allows the portfolio to drift with the trends of the market. It uses the minimum possible information from the data, but it explodes the trends to attain profit and avoid transactions costs.

The proposed method exploited information from three different sources. The first one is the market itself. The method considered transaction costs, unbalance, and inflation. All of them are related to properties of the market. Therefore, the method considered information from the market's nature into the optimization. The second source is price data, which is the same one used by the single-period method. Nevertheless, the investment strategies method uses it to make estimations further into the future and for evaluation of the current state of the portfolio (i.e., data innovations). Finally, it takes advantage of the market trends like the buy-and-holds do. The time horizon of the portfolio is an estimation about the market trend. Therefore, information from trends is also exploited by the investment strategies method.

The conclusion from this analysis is that information is crucial to make good investment decisions, and it is better to use information from the maximum possible number of sources. The poor performance of the single-period method could be attributed to it considering information from one source only. In other words, it used all the information from the price data but ignored the current state of the market and its trends.

The results showed cases of both relatively extraordinary loss and some scarce cases of relatively extraordinary profit (compared to the average value). Nonetheless, these fortunate cases seemed to be random and non-predictable, and therefore, it would not be possible to devise a strategy to exploit them. This result favors the efficient market hypothesis, although the experiments stopped when the portfolio reached the goal; therefore, there is no way to determine whether the results truly showed extraordinary profit. This is still an open question.

The technical approach is based on the idea that data history holds all the information about a security price, but the experiments indicated some scenarios were not predictable. There are two possible explanations: These scenarios could be predicted from other information sources besides data history. This would contradict the technical approach. The second one is that these scenarios are predictable with a further analysis of the data. The present method did not consider volume information or any

technical indicators. This matter needs to be further investigated.

The decisions made by the method could be described to be conservative. The goal was set based on the market conditions and the investor's preference. In the original algorithm, evidence is provided to the investor to help him determine a reasonable goal. If the investor decides the return's goal is not enough for him, this means he should be willing to take higher levels of risk to attain it. In other words, the method only provides goals which are attainable given the current market conditions. The method could be applied several times to accumulate any desired level of wealth. This approach appears to indicate that profit can only be attained with time and effort. Nevertheless, the case where the portfolio's value is higher than the goal was not fully investigated in the experiments. Further investigation is needed to arrive at conclusions about this favorable scenario and its relationship with final return. New approaches could be devised from that analysis.

Finally, the investment strategies method allowed emulation of different behaviors. For example, when the investment strategy chooses portfolios based on their returns, it behaves like a Martingale strategy: It will choose the highest return portfolios when the current wealth is less than the goal, but it will choose the lowest risk portfolios in the contrary case. The opposite occurs when risk is the only factor considered by the investment strategy. The behaviors observed by the method are not that different from behaviors observed in investors, but the difference in performance would come from the information available to the method and its use of it. The conclusion is that a good investment method should be wise enough to decide when to take action and when to let itself to "sail freely with the wind of change."

The method is still open to improvement. It could include other sources of information (volume, news, and others). The experiments showed more information is better when making decisions, but it should be used carefully. Also, static restrictions could be included as well as trading execution capabilities. This problem is concerned with methods to execute the trading orders, which could be impossible given the market conditions. Moreover, a high volume of transactions of the same asset may have an impact on its price [2].

Another improvement is the use of other techniques to model the future behavior of returns. A multivariate distribution was used in this work to simulate different scenarios and estimate the distribution of the individuals. Finance theory has studied GARCH processes to model the volatility of time-series. A combination of GARCH with evolutionary algorithms could be the next stage of the proposed system.

Finally, this work considered the investment strategy was fixed along the run. It was shown how the behavior of the system changes with the selected strategy. Therefore, there could be cases where a particular strategy is better than others. A method to update strategies dynamically is open to future research.

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Compliance with ethical standards

Conflict of interest The authors declare they have no conflict of interest.

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